Bio 338 Modeling Biological Dynamics Notes

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1 Basic Modeling with ODEs

1.1 Introduction

State: numbers that describe variables (state variables) of interest
State variables: Things we care about and measure
State space: space of allowable values of a state variable
One thing to note is that units can affect the state space. For example, the state space for allowable temperatures in Celsius is $C \in [-273.15, +\infty)$ whereas the allowable temperatures in Kelvin is $K \in [0, +\infty)$.
We can express differential equations as compartment diagrams. This diagram represents linear growth and is

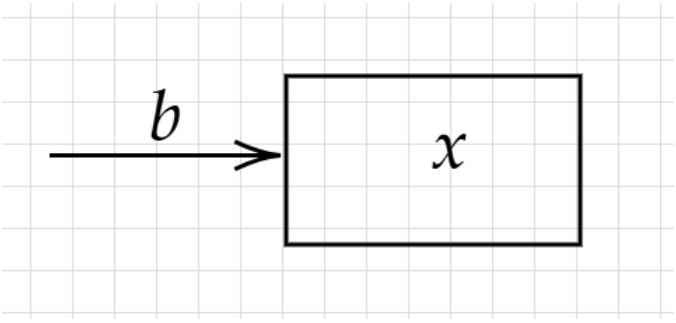


Figure 1: Compartment diagram of linear growth

isomorphic to the equation

$$\dot{x} = b$$

For these examples, let us use x to describe the rabbit population. In this case, the population is increasing by b rabbits per time unit (we shall say month for the example). Figure 2 is the compartment diagram for the equation

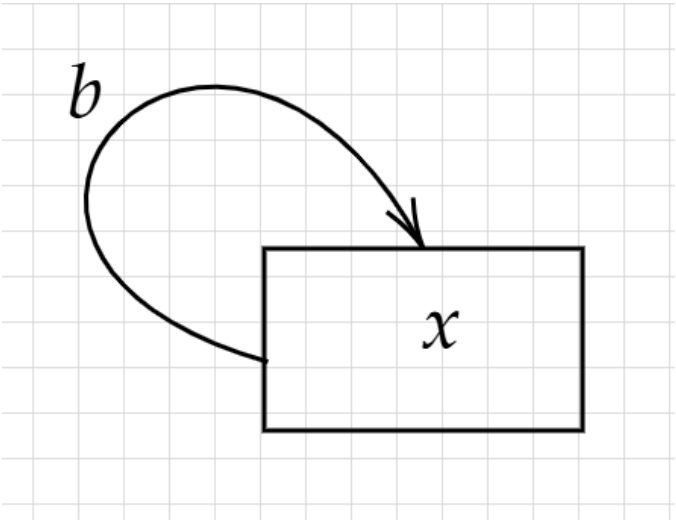


Figure 2: Compartment diagram of per capita growth

$$\dot{x} = bx$$

This effectively represents that more rabbits means more offspring which physically makes sense. We can adjust this model further by adding a death rate d. We can then draw the compartment diagram as seen in Figure 3. The isomorphic differential equation for this is

$$\dot{x} = bx - dx = (b - d)x$$

which has the solution

$$x(t) = x_0 e^{rt} \qquad r = b - d$$

If $r > 0$, then the birth rate "out competes" the death rate which leads to exponential growth. On the other hand, if $r < 0$, the death rate overrides the birth and hence the rabbit population disappears. We can make it more realistic by adding carrying capacity that only allows k rabbits in our system. This system is described by the *Euler-Lotka* Model

$$\dot{x} = r\left(1 - \frac{x}{k}\right)x$$

where r is once again the sum of the birth and death rates. The second term effectively scales the birth and death based off of how close the population is to the carrying capacity. We must have that $r > 0$ or else we see that the amount of deaths slows down as we approach the carrying capacity which is unphysical. If we think about when $r < 0$, we also see that if $x > k$, we have unbounded growth since r and $\left(1 - \frac{x}{k}\right)$ is also negative leading to $\dot{x} > 0$ and hence unbounded growth. Even if it hypothetically did work, we wouldn't really care since $r < 0$ should force everything to zero and so k doesn't have any impact on our model. Checking the behavior when $r > 0$, we find that if $0 < x < k$, we should expect growth and indeed $\dot{x} > 0$. When the number of rabbits is above the carrying capacity, we should expect the population to decline and indeed that occurs as $\dot{x} < 0$.

It is also worth noting that this model assumes the number of rabbits is a continuous variable which implies fractional (or irrational) number of rabbits which is clearly not reality. We disregard this concern for large populations because the difference between 1000.5 and 1001 rabbits is trivial; however, for small populations, we would need to enforce a discrete model since the difference of one rabbit is more significant on that scale.

1.2 Predation

Suppose that foxes (y) eat rabbits (x). Also suppose that the rate of foxes is dependent on the food supply. We can then write the differential equation describing the number of foxes as

$$\dot{y} = b[\text{food}]y - Dy$$

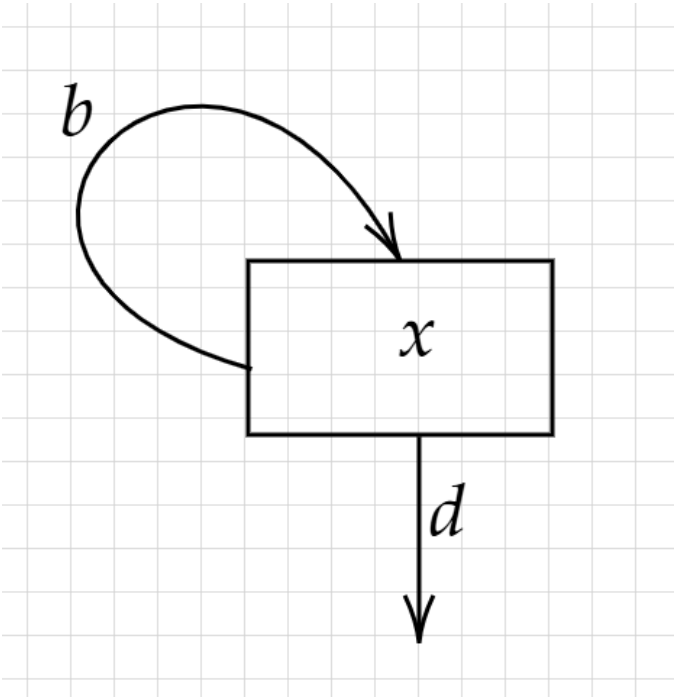


Figure 3: Compartment diagram of per capita growth and death

We can define the food supply as the probability of a single predation event as p multiplied by the number of rabbits there are x. We can then multiply the food supply available by the number of wolves to feed this which hence leads to

$$\dot{y} = bpxy - Dy$$

where D is the death rate per capita for wolves. Before, we described the rabbits as just $\dot{x} = rx$. Now we have to factor in the deaths due to foxes which is once again the food supply multiplied by the number of foxes. We now have the set of differential equations

$$\dot{x} = rx - pxy$$
$$\dot{y} = bpxy - Dy$$

And if we let all the coefficients equal 1, we get the *Lotka-Volterra Model*.

$$\dot{x} = x - xy$$
$$\dot{y} = xy - y$$

1.3 HIV

What perplexed people about HIV was the timeline of the disease. The viral load would initially spike and decline within the first 8 weeks of infection. It would then stay at a low viral load for years (in what is deemed its latent phase) before slowly increasing and leading to AIDS. There were two main questions:

1. What causes the initial spike and decline?
2. What causes the sudden climb in viral load leading to AIDS?

The initial hypothesis was that there wold be rapid viral growth and then the immune system would respond (leading to increase and decline in viral load) before replicating slowly in the background. In 1994, the first antiretroviral drugs were in clinical trials. These vaccines worked by stopping viral replication rather than directly killing existing viral particles. According to the previous hypothesis, these drugs shouldn't make a significant impact since the viruses are only replicating slowly. What they noticed was an exponential decay in viral load as time progress after the treatment.

In 1996, Alan Perelson publishes *HIV-1 Dynamics in Vivo: Virion Clearance Rate, Infected Cell Life-Span, and*

Viral Generation Time to explain what was observed. Let us consider a basic bathtub with a faucet and a drain. The inflow is described as Q_i and the outflow is described as Q_o and is dependent on the volume in the bathtub. We can therefore describe the amount of volume in the bathtub, V as the following differential equation

$$\dot{V} = Q_i - Q_o V$$

We especially care about the steady state condition which is when $\dot{V} = 0$. This leads to $V = \frac{Q_i}{Q_o}$ which effectively showed that the volume at steady state is dependent on the *ratio* and not the magnitudes of Q_i and Q_o. We could therefore have a faucet with high mass flow rate but also a really efficient drain or a dripping faucet and inefficient drain. We can't tell by solely looking at the amount of water in the drain. This extremely basic model already gives the insight that the virus is replicating extremely quickly, but the immune system is also clearing them quickly too. If we solve for $V(t)$ generally, we get

$$\begin{aligned} \dot{V} &= Q_i - Q_o V \\ \int \frac{1}{Q_i - Q_o V} dV &= t + c \\ V(t) &= \frac{Q_i}{Q_o} + Ce^{-Q_o t} \end{aligned}$$

If we apply the initial condition that $V(t=0) = V_0$, then we find the constant to be $C = V_0 - \frac{Q_i}{Q_o}$. If Q_i and Q_o are small, we should expect small exponential decay and vice versa. This effectively explains the dramatic difference the antiretrovirals made. Let us now consider a more sophisticated model. One thing to note is that the line from N_v

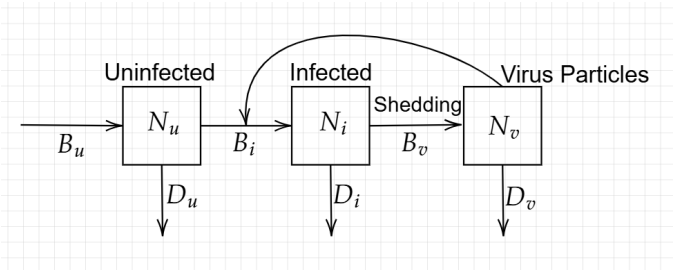


Figure 4: Compartment diagram of a model similar to Perelson's

implies that the rate of infection is dependent on the number of virus particles. There is also a negligible consumption of N_v in this process. Shedding also depends on N_i but doesn't change N_i. We can then write these as a set of differential equations

$$\begin{aligned} \dot{N}_u &= B_u - D_u N_u - B_i N_v N_u \\ \dot{N}_i &= B_i N_v N_u - D_i N_i \\ \dot{N}_v &= B_v N_i - D_v N_v \end{aligned}$$

These equations are annoying to solve analytically but we can solve it numerically. We can then graph as seen in Figure 5. We can further adjust the model to include latent infected cells: that is, cells that are infected but do not shed viruses. Let us say that 10% of cells become latently infected. Let us assume the death rate of latent cells is the same as uninfected ($D_l = D_u$). Let us also say that the latent cells become active (and begin shedding) at a rate B_a. Our set of differential equations is now

$$\begin{aligned} \dot{N}_u &= B_u - D_u N_u - B_i N_v N_u \\ \dot{N}_i &= .9 B_i N_v N_u + B_a N_l - D_i N_i \\ \dot{N}_l &= .1 B_i N_v N_u - B_a N_l - D_u N_l \\ \dot{N}_v &= B_v N_i - N_v D_v \end{aligned}$$

1.4 Tangent Spaces

Definition 1.1 (Tangent Space)**.** *The tangent space of a* manifold *is a vector space that contains all possible directions tangent to the manifold at that point.*

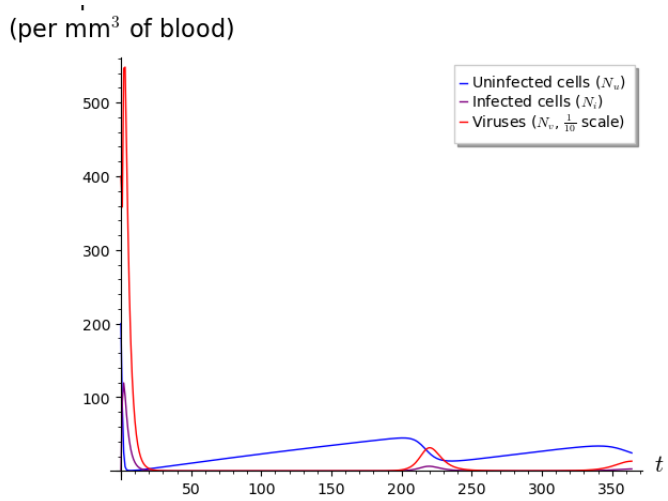


Figure 5: Plot of the amount of HIV in the blood over time with the parameters using McClain's model: $B_u = 0.272, B_i = 0.00027, B_v = 100, D_u = 0.00136, D_i = 0.33, D_v = 2$

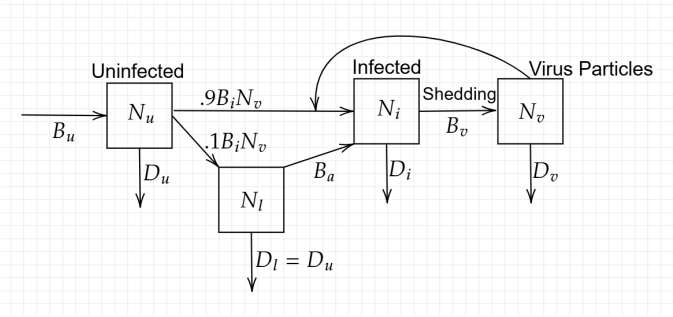


Figure 6: McClain's model including latently infected cells

Definition 1.2 (Manifold)**.** *An n-dimensional manifold is a topological space with the property that each point has a neighborhood that is homeomorphic to an open subset of n-dimensional Euclidean space.*

We can effectively think of the a state space as a manifold and each point on that manifold corresponds to a specific state in that state space. We can then think of the tangent spaces as all possible changes that can be made to a certain state in that state space. I wish I knew more about differential geometry but maybe Gábor Székelyhidi can teach me about it.

Definition 1.3 (Vector Field)**.** *A vector is a function f : points in the state space* $(\mathbb{R}^n) \mapsto$ *points in the tangent space* $(\mathbb{R}^n)$*.*

Let us look at the example $x \in [0, \infty)$ to be our state space and $\dot{x} = 2x$ as our manifold. The state space is the positive half line and so vectors in this tangent space can point in either direction. So -5 or 2 may be allowable vectors in this tangent space. However, the *vector field* is strictly enforced by $\dot{x} = 2x$ which is a *subset* of the tangent space. We can also look at the vector field for the logistic model where $x \in [0, \infty)$ and $\dot{x} = rx\left(1 - \frac{x}{k}\right)$, $r > 0$. We now note for $x < k$, the vector field points in the $+\hat{x}$ whereas if $x > k$, the vector field points in the $-\hat{x}$ direction. That is because there exists a fixed point at k.

Theorem 1.4 (Banach Fixed Point Theorem)**.** *Let* (X, d) *be a* complete *metric space with a contraction mapping* $T : X \mapsto X$*. Then T admits a unique fixed point* $x^* \in X$ *meaning* $T(x^*) = x^*$*. Furthermore,* X^* *can be found as follows: start with an arbitrary element* $x_0 \in X$ *and define* $x_n = T(x_{n-1})$ *for* $n \in \mathbb{N}$*. Then,* $\lim_{x\to\infty} x_n = x^*$

Let us look at some vector fields in $\mathbb{R}^2$. Consider the Romeo and Juliet system of differential equations which models Romeo's love y and Juliet's love x as

$$\dot{y} = -x$$
$$\dot{x} = y$$

We can also take a look at the Lotka-Volterra model which describes predator-prey dynamics and is described by

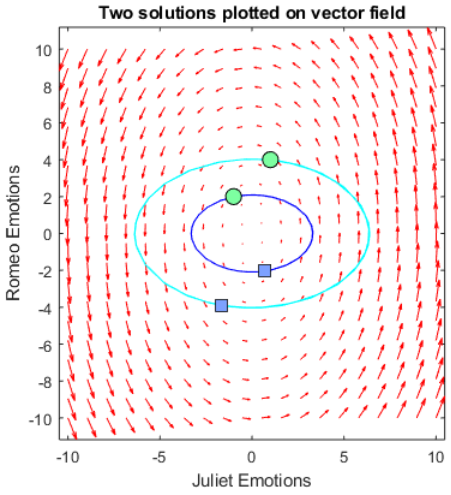


Figure 7: Vector field of the Romeo and Juliet set of differential equations

the equations

$$\dot{x} = x - kxy$$
$$\dot{y} = kxy - y$$

We can see that in Figure 8, there exists a fixed point at the origin (no predators and no prey so no dynamics) and another one somewhere in the positive plane which suggests the existence of a steady state.

Definition 1.5. Trajectories *are solutions to ODEs that exist and are unique (assuming the conditions of the Picard-Lindelöf Theorem are satisfied) and are* curves *in the state space that follow the vector field.*

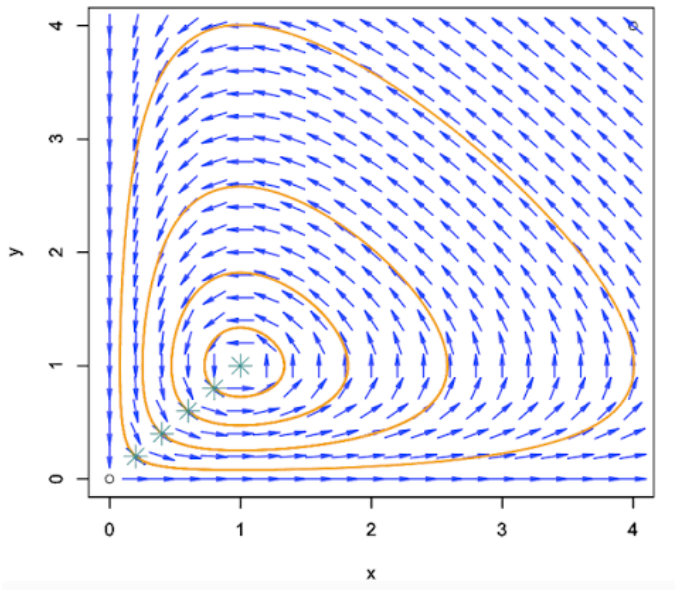


Figure 8: Vector field of the Lotka-Volterra model

We can describe a vector field V to be a function $V : \mathbb{R}^n \mapsto T_pM$ where T_pM denotes the tangent space at a point p along the state space manifold M. It is worth noting that two trajectories can *never* cross since that would require the vector field to be multi valued which is not possible over $\mathbb{R}$. One way to approximate solution curves to ODEs is via Euler's method which is as follows

$$x(t + \Delta t) = x(t) + x'(t)\Delta t$$

Other numerical methods improve on Euler's method like Runge-Kutta but Euler's method is the most clear to see. Some numerical integrators can have adaptive time steps that are based on the magnitude of the local vector field. This is because small time steps offer better accuracy at the expense of compute time. Large time steps are faster to calculate but may not converge to the true solution and are prone to deviations.

1.5 Fixed Points and Stability

Definition 1.6. *A* fixed point $\mathbf{x}$ *is a point in the state space such that* $\dot{\mathbf{x}} = 0$.

The vector field at a fixed point is the zero vector. We can classify the stability of a fixed point into a few categories:

(i) A system is *stable* if it returns to its fixed point when given a slight perturbation from it

(ii) A system is *unstable* if it leaves its fixed point when given a slight perturbation

(iii) A system is *quasi-stable* or *semi-stable* if one direction is stable while the direction is unstable (saddle point). We generally consider this to be unstable since there is only an infinitesimally small line of stability and the rest of the space unstable.

It is worth noting that there are different notions of stability. We shall state two.

Definition 1.7. *Consider an autonomous nonlinear dynamical system*

$$\dot{x} = f(x(t)) \qquad x(0) = x_0$$

where $x(t) \in \mathcal{D} \subseteq \mathbb{R}^n$ *denotes the system state vector,* $\mathcal{D}$ *denotes an open set containing the origin, and* $f : \mathcal{D} \mapsto \mathbb{R}^n$ *is a continous vector field on* $\mathcal{D}$*. Suppose* f *has an equilibrium at* x_e *such that* $f(x_e) = 0$*. Then:*

(i) The fixed point is said to be Lyapunov stable *if for every* $\epsilon > 0$ $\exists \delta > 0$ *such that* $||x(0) - x_e|| < \delta$ *then for every* $t \geq 0$ *we have that* $||x(t) - x_e|| \leq \epsilon$*.*

(ii) The fixed point is said to be asymptotically stable *if it is Lyapunov stable and there exists a* $\delta > 0$ *such that if* $||x(0) - x_e|| < \delta$ *then* $\lim_{t\to\infty} ||x(t) - x_e|| = 0$

In essence, a *Lyapunov stable* system just needs to stay within an ϵ distance from the fixed point given it starts δ away. An *asymptotically stable* system implies that close enough starting conditions will lead to convergence to the fixed. This means something that just orbits the fixed point can be consider Lyapunov stable but not asymptotically stable.

Definition 1.8. *A* nullcline *is a curve in the tangent space where the vector field is 0 in a particular direction.*

In a two dimensional state space, the location where two nullclines meet is a fixed point provided the nullcines are of *different* state variables. If we look at the Lotka-Volterra models and add in competition or communicable disease, we have

$$\dot{x} = Ax - Bxy - x^2$$
$$\dot{y} = Cy - Dxy - y^2$$

We can solve for nullclines by setting $\dot{x} = \dot{y} = 0$ and solving a system of equations. If we ignore the trivial nullclines, we get that nontrivial fixed point of this model is

$$x = \frac{A - BC}{1 - BD} \qquad y = C - D\left(\frac{A - BC}{1 - BD}\right)$$

When $\frac{A}{B} > C$, this fixed point is stable but when $\frac{A}{B} < C$, this fixed point is now unstable.

Definition 1.9. *An* attactor *of a dynamical system is a set of states to which the system tends to evolve for a large number of starting states. We define "a large number" to be a set that isn't of measure* 0*.*

An attractor can be a stable fixed point or a limit cycle. It cannot be an unstable fixed point nor a saddle (which is a set of starting states that leads to a fixed point of measure 0). The *Basin of Attraction* is the set of states that will evolve to a particular attractor over time. Let us consider a logistic model with the allee effect which states that a population needs to be above some minimum value to reproduce. It is described by the following model:

$$\dot{x} = rx\left(1 - \frac{x}{k}\right)\left(\frac{x}{a} - 1\right) \qquad 0 < a < k$$

We see that there are fixed points at $x = 0, a, k$. If we take derivatives of the time deriavtives with respect to x we can classify their stabilities

$$\frac{\partial \dot{x}}{\partial x} = r\left(1 - \frac{x}{k}\right)\left(\frac{x}{a} - 1\right) - \frac{r}{k}x\left(\frac{x}{a} - 1\right) + \frac{r}{a}x\left(1 - \frac{x}{k}\right)$$

At $x = a$, we have

$$\left.\frac{\partial \dot{x}}{\partial x}\right|_{x=a} = r\left(1 - \frac{a}{k}\right) > 0$$

which therefore makes this point unstable. At $x = k$, we have

$$\left.\frac{\partial \dot{x}}{\partial x}\right|_{x=k} = r\left(\frac{k}{a} - 1\right) < 0$$

and is hence stable. At $x = 0$, the derivative evaluates to $-r < 0$ and is also stable. We therefore have two basis of attraction: one that goes towards $x = 0$ and one towards $x = k$. The set $[0, a)$ leads to the fixed point $x^* = 0$ and the set $(a, +\infty)$ leads to $x^* = k$. An attractor is an element of its basin. The boundary between basis is known as the *separatrix*. In one dimension, it is an unstable fixed point. In two dimensions, its a curve. For higher dimensions, the separatrix is always a hyperplane.

1.6 Bifurcations

Recall in the competitive Lotka-Volterra model that depending on whether $\frac{A}{B} > C$ determined the stability of the fixed point. This transition due to a change in parameters is known as a *bifurcation*. Let us look at an example in one dimension. E. coli can consume lactose but importing lactose is an energy expensive process. As such, there is a switch that causes the E. coli to import lactose or not. Let us define x to be the intracellular lactose level. The change in it $\dot{x}$ is equal to the production minus the degradation. The production is modeled by

$$\text{production} = \frac{a + x^n}{1 + x^n}$$

and the degradation is per capita degradation and therefore modeled as rx. As such,

$$\dot{x} = \frac{a + x^n}{1 + x^n} - rx$$

We can plot the production and degradation terms separately to get Figure 9 Regions where the red line (degrada-

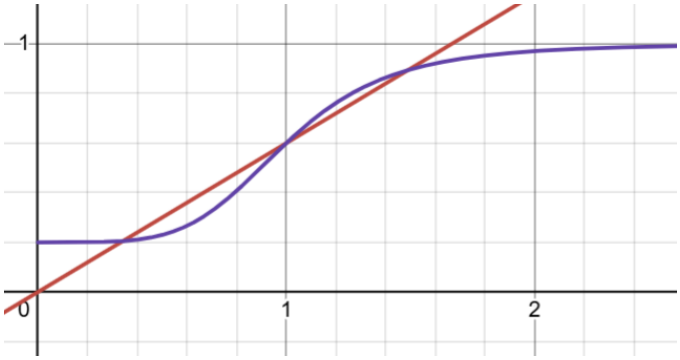


Figure 9: Production and degradation terms plotted separately via the over-under method

tion) is greater than the purple sigmoid (production) is net negative change in lactose levels and vice versa. For a small r, the red line in Figure 9 is very shallow and intersects the production term at one large x^*. As r increases, eventually we eventually hit a bifurcation as now the degradation intersects the production term at a lower x^* which then splits into a middle unstable fixed point and a lower stable fixed point. As r increases even more, the top two fixed points collide and annihilate each other and all that remains is the lower fixed point. These two bifurcations are known as a "saddle node" bifurcation.

In 1 dimension, we always require an unstable fixed point between two stable fixed points. This is easy to see because if we have some sort of smooth function γ that describes the value of $\dot{x}$ and in so the middle region $[x^*_-, x^*_+]$, we require $\gamma > 0$ or $\gamma < 0$ and it must remain that value within that interval by the Intermediate Value Theorem. We can generalize this to higher dimensions by requiring a separatrix.

Let us look at a two gene toggle which can be represented by x inhibits the production production of y and vice versa.

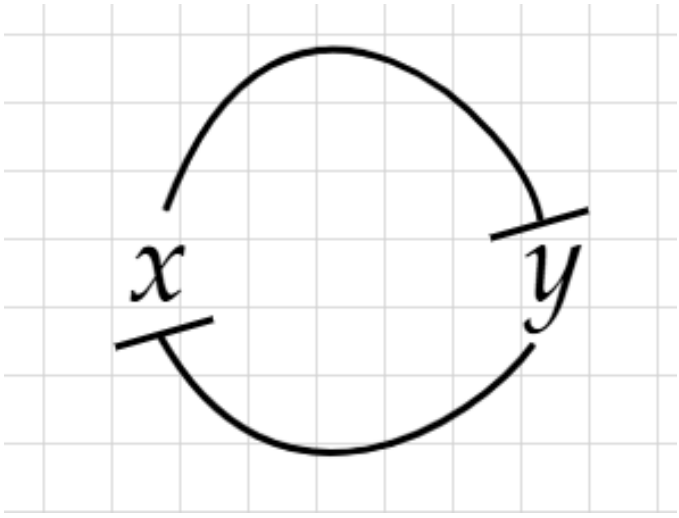


Figure 10: Diagram to describe the relationship of this two gene toggle

In other words, they mutually inhibit one another. Repression of either gene requires binding of the repressor to the other genes operon. If, for example, we take $\mathbb{P}(y \text{ binds operon } x)$, we can express the dynamics of this probability as

$$\begin{aligned}\dot{\mathbb{P}}(\text{bound}) &= (\text{binding rate})\mathbb{P}(\text{unbound}) - (\text{unbinding rate})\mathbb{P}(\text{bound}) \\ &= k_{\text{on}}C(1 - \mathbb{P}(\text{bound})) - \beta_{\text{off}}\mathbb{P}(\text{bound})\end{aligned}$$

where k_{on} is the binding rate per repressor (mM), C is the concentration of repressor (mM), and β_{off} is the dissociation rate. In steady state, we have that $\dot{\mathbb{P}}(\text{bound})= 0$. In that case, we have

$$k_{\text{on}}C(1 - \mathbb{P}(\text{bound})) = \beta_{\text{off}}\mathbb{P}(\text{bound})$$
$$¶(\text{bound}) = \frac{k_{\text{on}}C}{\beta_{\text{off}} + k_{\text{on}}C}$$

which is the location of this fixed point. Doing some more algebraic manipulation, we get that

$$\begin{aligned}\mathbb{P}(\text{bound}) &= \left(1 + \frac{\beta_{\text{off}}}{k_{\text{on}}C}\right)^{-1} \\ &= \frac{1}{1 + \frac{K_d}{C}}\end{aligned}$$

where K_d is the equilibrium dissociation constant and described by $\frac{\beta_{\text{off}}}{k_{\text{on}}}$. Since the cumulative probability of the repressor being bound and unbound is 1, we can write that

$$\begin{aligned}\mathbb{P}(\text{unbound}) &= 1 - \frac{1}{1 + \frac{K_d}{C}} \\ &= \frac{1}{\frac{C}{K_d} + 1}\end{aligned}$$

Sometimes we want to incorporate some sort of cooperativity where we require multiple repressors to bind to an operon. n can be non-integer values because sometimes one can get away with repression with different amounts of repressors. If we let RNA production rate be a function of the repressor concentration c, we have that

$$f(c) = \mathbb{P}(\text{unbound})\Gamma$$

where Γ is the maximum production rate. If we add in a per capita clearance rate, we can describe this two switch gene toggle as

$$\dot{x} = \frac{\Gamma_x}{\left(\frac{y}{K_{d,y}}\right)^n + 1} - k_x x$$
$$\dot{y} = \frac{\Gamma_y}{\left(\frac{x}{K_{d,x}}\right)^n + 1} - k_y y$$

1.7 Oscillations

Historically, we thought that we need an external driving force for self sustaining oscillations. This isn't true. Instead, autonomous oscillations require some form of negative feedback. This is a necessary but not a sufficient condition for oscillations. The following cannot have oscillations since both feedback are positive. The configuration in Figure 10 is also incapable of support oscillations since a negative times a negative is positive and hence no oscillations. If x promotes y but y antagonizes x, oscillations are possible. However, there also needs to be a delay for oscillations to occur. There are two types of oscillations: transient and limit cycle. Transient oscillations settle down to a fixed point. Limit cycles are sustained oscillations that don't settle down towards a fixed point. Limit cycles can be described as Lyapunov stable attractors and transient oscillations are most specifically described as asymptotically stable attractors. If we recall the Romeo and Juliet model, $\forall r \in \mathbb{R}$ where $r = \sqrt{x^2 + y^2}$, there exists a stable limit cycle where the amplitude is r which is dependent on the initial conditions. In the Lotka-Volterra model, the period and amplitude of the oscillations are also dependent on the initial conditions. These are undesirable behaviors when we want to regulate these oscillations. Three element oscillators help overcome these limitations.

Let us take a look at this model that describes hormonal cycles. The hypothalamus secretes gonadotropin-releasing hormone (GnRH) which promotes the pituitary gland to secrete luteinizing hormone (LH) and folice stimulating hormone (FSH) which then promotes the ovaries to secrete estrogen and progesterone which downregulates GnRH production. This is a negative feedback system since we have two positive terms and one negative. Downregulation can either be increased degradation or decreased production. In this case, the ovaries decrease the production of

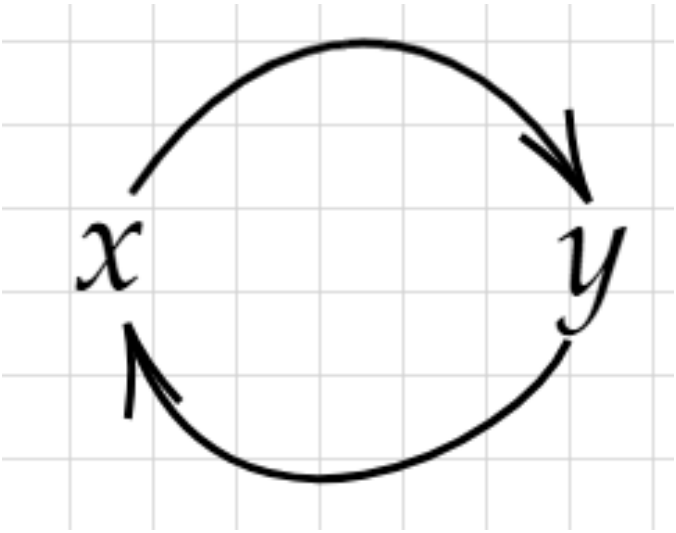


Figure 11: Both are positive and so no oscillations

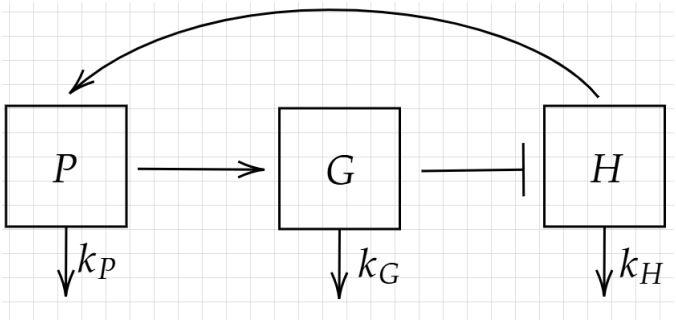


Figure 12: Model of hormonal cycles due to the pituitary gland, hypothalamus, and the ovaries

GnRH by the hypothalamus. If we use the double suppressor function to decsribe G's effect on H and let $\Gamma = 1$ and $K_d = 1$, we get

$$\begin{aligned}\dot{H} &= \frac{1}{1+G^n} - k_H H \\ \dot{P} &= H - k_P P \\ \dot{G} &= P - k_G G\end{aligned}$$

For simplicity, we shall take the one constant approach which is $k_H = k_P = k_G = k$ for some random constant k. We can modify this model further by having a constant production term like

$$\dot{G} = P - kG + q$$

In general, we require either strong feedback with shorter time delay or weaker feedback with a longer time delay or some sort of combination in between. In general, there exists some separatrix that separates non-oscillatory and oscillatory behavior. Crossing this separatrix is a bifurcation. One of the most common types of these bifurcation is the *Hopf* bifurcation where a fixed point loses/gains stability that leads to a loss/gain of a limit cycle.

1.8 SIR Model

In around 1916, Ronald Ross classified three different types of disease dynamics:

- Endemic: low steady prevalence (tuberculosis, leprosy)
- Epidemic: intense outbreaks before disappearing (plague, cholera)
- Outbreaks: constantly present with regular flareups (measles, dysentery)

The question is then why are there these different dynamics? In 1927, Karmac and Kundric publish the SIR model. The constants β and γ are the transmission coefficient and recovery rates respectively. The model makes a few

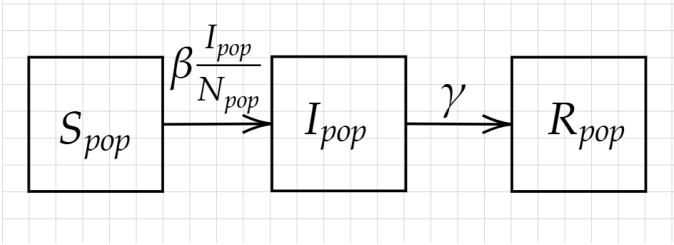


Figure 13: Compartment diagram of the SIR model

assumptions. The first is that once you are infected, you have complete immunity. Additionally, these dynamics are happening much quicker than the birth or death rates. Finally, everyone is equally susceptible and nobody is actually dying. We can express this in terms of differential equations as

$$\begin{aligned}\dot{S}_{pop} &= -\beta \frac{I_{pop}}{N_{pop}} S_{pop} \\ \dot{I}_{pop} &= \beta \frac{I_{pop}}{N_{pop}} S_{pop} - \gamma I_{pop} \\ \dot{R}_{pop} &= \gamma I_{pop}\end{aligned}$$

with initial conditions $I_{pop}(0) > 0, R_{pop}(0) = 0, S_{pop}(0) = N_{pop} - I_{pop}(0)$ where N_{pop} is the total population and $\frac{I_{pop}}{N_{pop}}$ is effectively the probability of encountering an infected person. The units of γ is reciprocal time and therefore $\frac{1}{\gamma}$ is effectively how long the infection lasts. β can be represented as $\dfrac{\#\text{ new infections}}{(\#\text{ infected})(\text{time})}$ and so therefore the ratio $R_0 = \frac{\beta}{\gamma} = \dfrac{\#\text{ new infections}}{\#\text{ infected}}$ and effectively represents the number of people a single infected person will infect. This

is known as the *basic reproduction ratio*. We want to know when an outbreak will arise which implies that $\dot{I}_{pop} > 0$ and therefore

$$\begin{aligned} \beta \frac{I_{pop}}{N_{pop}} S_{pop} - \gamma I_{pop} &> 0 \\ \beta \frac{S_{pop}}{N_{pop}} &> \gamma \\ \frac{S_{pop}}{N_{pop}} &> \frac{1}{R_0} \end{aligned}$$

which tells us that when the proportion of susceptible people is larger than the reciprocal of the reproduction rate, an outbreak will occur. If R_0 is very large, and therefore very transmissible, it is very easy to have an outbreak. To do any further analysis, let us non-dimensionalize using

$$S = \frac{S_{pop}}{N_{pop}} \qquad I = \frac{I_{pop}}{N_{pop}} \qquad R = \frac{R_{pop}}{N_{pop}}$$

This leads to

$$\dot{S} = \frac{d}{dt}\left(\frac{S_{pop}}{N_{pop}}\right) = \frac{1}{N_{pop}}\left(-\beta \frac{I_{pop}}{N_{pop}} S_{pop}\right) = -\beta IS$$

so therefore our full system of differential equations is

$$\begin{aligned} \dot{S} &= -\beta IS \\ \dot{I} &= \beta IS - \gamma I \\ \dot{R} &= \gamma I \end{aligned}$$

We clearly see there is a fixed point at $I = 0$ but we have no notions of what S and R are at this fixed point. If we look at the time series, we expect that

$$\lim_{t\to\infty} S = 1 - \lim_{t\to\infty} R$$

We can't solve things analytically but we can abuse the Leibniz notation

$$\frac{dS}{dR} = \frac{\frac{dS}{dt}}{\frac{dR}{dt}} = \frac{\dot{S}}{\dot{R}} = \frac{-\beta IS}{\gamma I} = -R_0 S$$

which therefore we get the solution to be

$$S(R) = Ae^{-R_0 R(t)}$$

where $A = S(R = 0) = S(t = 0) \sim 1$. We therefore have

$$\dot{R} = \gamma I = \gamma(1 - S - R) = \gamma(1 - e^{-R_0 R} - R)$$

As we let $t \to \infty$, R should go to some constant C so $\dot{R} = 0$ and therefore $R = 1 - e^{-R_0 R}$. We also note that system is two dimensional since $S + I + R = 1 \to \dot{S} + \dot{I} + \dot{R} = 0$ and so we can always rewrite one variable in terms of the other two. We find the nullclines to be $S = 0, I = 0$ for the S-nullclines and $I = 0, S = \frac{1}{R_0}$ for the I-nullclines.

1.8.1 SIRS-Waning Immunity

We can adjust the model by implementing waning immunity which is conveyed via some constant α. In this case, our system of differential equations is

$$\begin{aligned} \dot{S} &= \alpha R - \beta IS \\ \dot{I} &= \beta IS - \gamma I \\ \dot{R} &= \gamma I - \alpha R \end{aligned}$$

The system is still closed so we have $S + I + R = 1$. If we rewrite R in terms of S and I, then we can rewrite the above to find the nullclines

$$\begin{aligned} \beta SI &= \alpha - \alpha S - \alpha I \\ I(\beta S + \alpha) &= \alpha(1 - S) \\ I &= \frac{\alpha(1_S)}{\beta S + \alpha} \end{aligned}$$

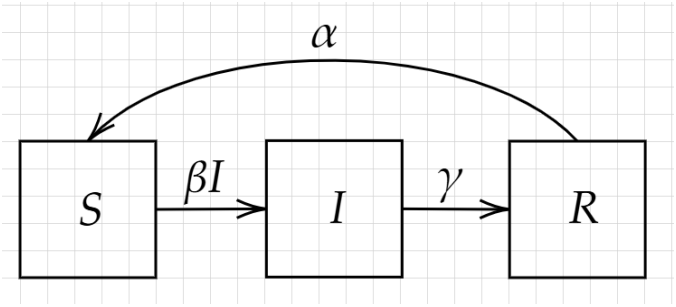


Figure 14: SIR model with waning immunity

which passes through the points $(1, 0)$ and $(0, 1)$ in the SI-plane. For the I nullcline, we have $S = \frac{1}{R_0}$. Therefore if we have $\frac{1}{R_0} < 1$, we have an *endemic fixed point*. This is where there are non zero infections yet the state of the system isn't changing. When $\frac{1}{R_0}$ becomes larger than 1, then the endemic fixed point disappears and the system undergoes a transcritical bifurcation. What's interesting to note here is that α does not affect if there *exists* an endemic fixed point but it does affect *how many* people are infected at the fixed point.

1.8.2 SIR with Births and Deaths

Now let us consider a system in which there is a constant birth rate and death rate per capita. The constant birth

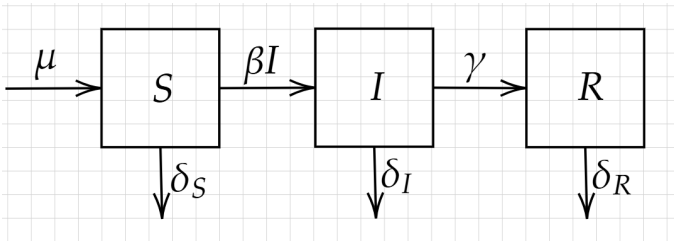


Figure 15: SIR model with births and deaths

rate is fine under large and stable population. We can describe the dynamics with

$$\begin{aligned}\dot{S} &= \mu - \beta IS - \delta_S S \\ \dot{I} &= \beta IS - \gamma I - \delta_I I \\ \dot{R} &= \gamma I - \delta_R R\end{aligned}$$

It is worth noting that this system is *not* closed and therefore our state variables are fractions of the *initial* population unless the births exactly balances the deaths. In other words,

$$\mu = \delta_S S(0) + \delta_I I(0) + \delta_R R(0)$$

For the S-nullcline, we get that $I = \frac{\mu - \delta_S S}{\beta S}$ and for the I-nullcline we have $I = 0$ or $S = \frac{\gamma + \delta_I}{\beta}$. This also has a transcritical bifurcation if $\frac{\gamma + \delta_I}{\beta}$.

1.8.3 SIRD - Deaths Explicitly

If we want to be more explicit with how we track the deaths, we can create another compartment. ρ is the probability of recovering instead of dying. This model is also closed. We can write the system as

$$\begin{aligned}\dot{S} &= -\beta IS + \alpha R \\ \dot{I} &= \beta IS - \gamma I \\ \dot{R} &= \rho\gamma I - \alpha R \\ \dot{D} &= (1 - \rho)\gamma I\end{aligned}$$

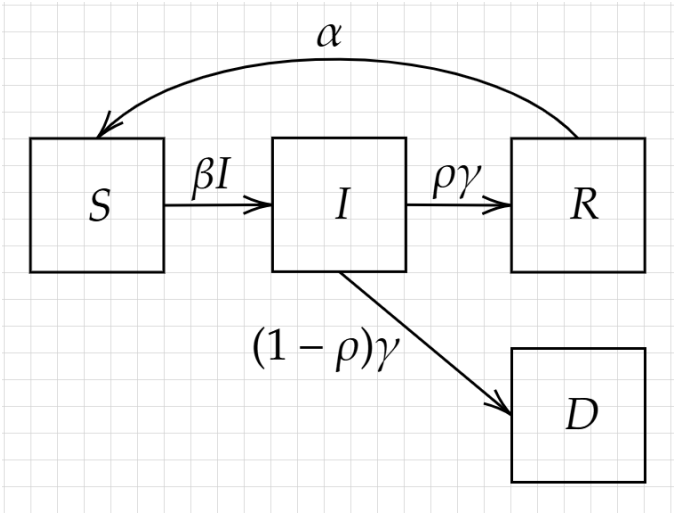


Figure 16: SIR model with an explicit death compartment

What we notice is that $\dot{D}$ can never be less than zero. What we can do is to attempt to minimize $\dot{D}$ which can be done by increasing ρ (analogous to better treatments), decreasing γ (keeping people sick instead of dying), or decreasing I (by reducing the rate of infection such that $\dot{I} \ll 0$). This implies that $\beta IS \ll \gamma I \to S \ll \frac{1}{R_0}$. We can do this by making S small, γ large (make duration shorter), or β small (reduce transmissibility).

2 Stochastic Processes

2.1 Probability

Chance is very important in many systems, especially when numbers are low. In higher populations, we can usually average over the probabilities. Let us start with some definitions.

Definition 2.1. *A* random variable X *is an observation of a random process.*

Definition 2.2. *A* random process *is a well defined process with an uncertain outcome.*

Definition 2.3. *A* sample space S *is the set of all possible outcomes for that process.*

Definition 2.4. *An event is a subset of outcomes defined in some way.*

The probability of an event A is denoted $\mathbb{P}(A)$. It is bounded on $[0, 1]$ and $\sum_{A \in S} \mathbb{P}(A) = 1$. The probability of events that aren't A is $\mathbb{P}(\text{ not } A) = \mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. The *joint probability* of two events A and B is $\mathbb{P}(A, B) := \mathbb{P}(A) \cap \mathbb{P}(B)$. The *conditional probability* of A given B is defined as $\mathbb{P}(A|B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$. We can say that $\mathbb{P}(A, B) = \mathbb{P}(B)\mathbb{P}(A|B) = \mathbb{P}(A)\mathbb{P}(B|A)$ which we can then derive Bayes' Theorem.

Theorem 2.5 (Bayes' Theorem)**.**

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(A)}{\mathbb{P}(B)}\mathbb{P}(B|A)$$

One example is a pregnancy test where $\mathbb{P}(-|XY)$ is high whereas $\mathbb{P}(XY|-)$ is not high. For other medical tests, we can use Bayes' Theorem to state

$$\mathbb{P}(\text{sick}|+) = \frac{\mathbb{P}(\text{sick})}{\mathbb{P}(+)}\mathbb{P}(+|\text{sick})$$

The first term is known as the positive predictive value (in Bayesian terms, the posterior probability of being sick) and is how likely you are really sick given a positive test. This is generally what you or the doctor care about when you take a medical test. $\mathbb{P}(\text{sick})$ is the prevalence (and prior belief of being sick) and the $\mathbb{P}(+|\text{sick})$ term is the sensitivity. In other words, if you're really sick, how likely is it to be accurate. This is the value that is put on the information pamphlet.

Definition 2.6. *A* probability distribution *is a function that gives the probability for all potential outcomes of a specific random process.*

In other words, there exists a functional form with parameters that dictate the probability for all outcomes (although the parameters may not be known to us). If $f(x)$ is a probability distribution, then $0 \leq f(x) \leq 1 \ \forall x$. We also have that $\sum_x f(x) = 1$ if $f(x)$ is discrete or $\int f(x)dx$ if $f(x)$ is continuous.

2.1.1 Binomial Distribution

The binomial distribution describes the probability of m success in n iid Bernoulli trials.

Definition 2.7. *An* iid *trial is one that is independent and identically distributed relative to other trials.*

Definition 2.8. *A* Bernoulli *trial is a random process with binary outcomes.*

We denote that a random variable X is distributed as a Bernoulli distribution with probability of success p as $X \sim \text{Ber}(p)$. The binomial distribution has two parameters: n being the number of trials and p being the probability of success for every trial. The random variable $m \sim \text{Bin}(n, p)$. Therefore we must have that

$$\mathbb{P}(m; n, p) = \mathbb{P}(m \text{ success})\mathbb{P}(n - m \text{ failures})(\text{number of permutations of } m \text{ successes and } n - m \text{ failures})$$

For m successes, that is equivalent as p^m and for $n - m$ failures, that is $(1 - p)^{n-m}$. The number of distinguishable permutations is given by ${}_nC_m = \frac{n!}{m!(n-m)!} = \binom{n}{m}$. Then $m \sim \text{Bin}(n, p) \to \mathbb{P}(m; n, p) = \frac{n!}{m!(n-m)!}p^m(1 - p)^{n-m}$. If we want the cumulative probability of say at least m successes, then we sum up all the probabilities for each $x \geq m$

$$\mathbb{P}(X > m) = \sum_{k=m}^{n} \binom{n}{k} p^k (1 - p)^{n-k}$$

We now introduce the *Method of Moments*. The method of moments allows to generate all the moments of a probability distribution which we can use to uniquely identify an unknown distribution.

Definition 2.9. *The* first moment *is the expected value denoted as $E(X)$ or $\langle X \rangle$. We can write it as*

$$E(X) := \sum_{x \in S} x\mathbb{P}(x) \ \textit{or} \ \int_S xf(x)dx$$

The expected value is also known as the population mean and it may not be an actual event in the sample space.

Definition 2.10. *The* second moment *is the variance and is defined as*

$$Var(x) := E((x - E(x)^2) = E(x^2) - (E(x))^2 = \int_S (x - E(x))^2 f(x)dx$$

The nth moment is defined as $E((x - E(x))^n)$. The third and fourth moments are known as the skew and kurtosis respectively. The skew measures how "lopsided" a distribution is whereas kurtosis measures how "pointy" a distribution is. To use the method of moments to find likely parameters to a function, we want to compare a moment with the sample value. For example,

$$\widehat{E(x)} = \bar{x}$$

where the hat denotes it being an estimate of the expected value/population and the bar on the RHS denotes the sample mean. Since the binomial distribution has two parameters, we got the second moment

$$\widehat{\text{Var}(x)} = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$$

2.2 Poisson Process

Suppose we have many trials $N \to \infty$ but each trial has a low probability of success. Considering dividing the N trials into n subsets small enough such that they contain at most one success. Then the probability of $\mathbb{P}(\text{success}) = p$. Each interval is effectively Bernoulli trial. We write $p = \frac{\lambda}{n}$. The total number of success as $m \sim \text{Bin}(n, p) = \text{Bin}(n, \frac{\lambda}{n})$. We want the limit $n \to \infty$ and so we have

$$\begin{aligned}
\lim_{n\to\infty} \binom{n}{m} p^m (1-p)^{n-m} &= \lim_{n\to\infty} \frac{n(n-1)\dots(n-m)\dots}{m!(n-m)(n-m-1)\dots} \left(\frac{\lambda}{n}\right)^m \left(1-\frac{\lambda}{n}\right)^{n-m} \\
&= \lim_{n\to\infty} \frac{n(n-1)(n-2)\dots(n-m+1)}{m!} \left(1-\frac{\lambda}{n}\right)^{n-m} \\
&= \frac{\lambda^m}{m!} \lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^n \left(1-\frac{\lambda}{n}\right)^{-m} \\
&= \frac{\lambda^m}{m!} \lim_{n\to\infty} \left(1-\frac{\lambda}{n}\right)^n \\
&= \frac{\lambda^m}{m!} e^{-\lambda}
\end{aligned}$$

Where $\frac{\lambda}{n}$ is very small so as $n \to \infty$, it goes to zero. We have now derived the Poisson Distribution which only depends on the parameter λ. One special thing about the Poisson distribution is that the mean and variance are the same.